

This file defines all 10 tables required for ElevatorIQ v1.1. Run schema_v1.sql against a PostgreSQL database to deploy. All tables, indexes, and immutability rules are included. Do not substitute SQLite or any other database.

Table Overview

Table	Role	Layer
customers	User accounts and plan tier	Application
cases	One analysis session per upload batch	Application
documents	Each uploaded file linked to a case	Application
reports	Generated PDF metadata and download tokens	Application
extractions_raw	Append-only audit trail of all structured JSON blobs	Flywheel — Immutable
facts_labor	Normalized labor rate data points for aggregation	Flywheel — Facts
facts_line_items	Normalized invoice line items with taxonomy tags	Flywheel — Facts
facts_parts	Parts and material cost data points	Flywheel — Facts
facts_contract_terms	Commercial and contract term data points	Flywheel — Facts
benchmarks	Versioned aggregated benchmark outputs	Flywheel — Intelligence

Deployment

TERMINAL

```
# Step 1: Create the database
createdb elevatoriq_dev

# Step 2: Run the schema
psql elevatoriq_dev < schema_v1.sql

# Step 3: Verify all 10 tables
psql elevatoriq_dev -c '\dt'
```

■ CRITICAL

Always use PostgreSQL. Do not substitute SQLite, MySQL, or any other database. Aggregation queries use PostgreSQL-specific functions (PERCENTILE_CONT, window functions).

customers

User accounts and plan tier

Column	Type	Null?	Notes
id	UUID	NO	Primary key. Default: gen_random_uuid()
email	TEXT	NO	Unique. Customer email address.
company	TEXT	YES	Optional. Company name.
plan_tier	TEXT	NO	Default: 'free'. Values: free pay_per_review portfolio
created_at	TIMESTAMPTZ	NO	Default: NOW()
updated_at	TIMESTAMPTZ	NO	Default: NOW()

SQL

```
CREATE TABLE customers (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email TEXT NOT NULL UNIQUE,
  company TEXT,
  plan_tier TEXT NOT NULL DEFAULT 'free',
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

cases

One analysis session per upload batch

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
customer_id	UUID	YES	FK → customers.id. Nullable for anonymous users.
review_type	TEXT	NO	invoice_review maintenance_bid_comparison modernization_comparison single_modernization contract_coverage
module	TEXT	NO	Derived: A B C
state	TEXT	YES	2-letter state code (e.g. MI). Used for benchmark lookup.
market	TEXT	YES	Metro market name (e.g. Detroit). Benchmark lookup.
equipment_type	TEXT	YES	hydraulic traction escalator other
status	TEXT	NO	Default: pending. Values: pending processing complete failed
customer_email	TEXT	YES	Where to send the PDF report.
created_at	TIMESTAMPTZ	NO	Default: NOW()
completed_at	TIMESTAMPTZ	YES	Set when status = complete.

SQL

```
CREATE TABLE cases (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  customer_id UUID REFERENCES customers(id),  
  review_type TEXT NOT NULL,  
  module TEXT NOT NULL,  
  state TEXT,  
  market TEXT,  
  equipment_type TEXT,  
  status TEXT NOT NULL DEFAULT 'pending',  
  customer_email TEXT,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  completed_at TIMESTAMPTZ  
);  
  
CREATE INDEX idx_cases_status ON cases(status);  
CREATE INDEX idx_cases_customer ON cases(customer_id);
```

documents

Each uploaded file linked to a case

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
case_id	UUID	NO	FK → cases.id. NOT NULL.
file_name	TEXT	NO	Original filename as uploaded.
file_type	TEXT	YES	User tag or auto-detected: contract invoice proposal callback_log equipment_list other
storage_path	TEXT	NO	S3 key or object path.
auto_detected	BOOLEAN	NO	TRUE if file_type was system-detected, not user-tagged.
uploaded_at	TIMESTAMPTZ	NO	Default: NOW()

SQL

```
CREATE TABLE documents (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  case_id UUID NOT NULL REFERENCES cases(id) ON DELETE CASCADE,  
  file_name TEXT NOT NULL,  
  file_type TEXT,  
  storage_path TEXT NOT NULL,  
  auto_detected BOOLEAN NOT NULL DEFAULT FALSE,  
  uploaded_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);  
  
CREATE INDEX idx_documents_case ON documents(case_id);
```

reports

Generated PDF metadata and download tokens

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
case_id	UUID	NO	FK → cases.id.
storage_path	TEXT	NO	S3 key for the generated PDF file.
download_token	UUID	NO	Unique token for secure download link. Expires.
token_expires_at	TIMESTAMPTZ	YES	Token expiry. Default: NOW() + 7 days.
emailed_at	TIMESTAMPTZ	YES	Timestamp when email was sent successfully.
created_at	TIMESTAMPTZ	NO	Default: NOW()

SQL

```
CREATE TABLE reports (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
case_id UUID NOT NULL REFERENCES cases(id),
storage_path TEXT NOT NULL,
download_token UUID NOT NULL DEFAULT gen_random_uuid(),
token_expires_at TIMESTAMPTZ DEFAULT NOW() + INTERVAL '7 days',
emailed_at TIMESTAMPTZ,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE UNIQUE INDEX idx_reports_token ON reports(download_token);
CREATE INDEX idx_reports_case ON reports(case_id);
```

extractions_raw

Append-only immutable audit trail

■ **CRITICAL**

This table is APPEND-ONLY. UPDATE and DELETE are blocked by database rules. Never modify existing rows.

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
case_id	UUID	NO	FK → cases.id.
module	TEXT	NO	A B C
state	TEXT	YES	2-letter state code. Anonymized geography.
market	TEXT	YES	Metro market name. No address or building name.
equipment_type	TEXT	YES	hydraulic traction escalator other
contract_type	TEXT	YES	full_maintenance oil_and_grease parts_and_labor modernization other
unit_count	INTEGER	YES	Number of elevator units in scope.
confidence_overall	TEXT	YES	high medium low
benchmark_version	TEXT	YES	Version of benchmarks used in this analysis.
raw_json	JSONB	NO	Full extraction JSON blob. Validated before insert.
created_at	TIMESTAMPTZ	NO	Default: NOW(). Immutable after insert.

SQL

```
CREATE TABLE extractions_raw (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  case_id UUID NOT NULL REFERENCES cases(id),  
  module TEXT NOT NULL,  
  state TEXT,  
  market TEXT,  
  equipment_type TEXT,  
  contract_type TEXT,  
  unit_count INTEGER,  
  confidence_overall TEXT,  
  benchmark_version TEXT,  
  raw_json JSONB NOT NULL,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);  
  
-- Immutability rules  
CREATE RULE no_update_extractions AS  
ON UPDATE TO extractions_raw DO INSTEAD NOTHING;
```

```
CREATE RULE no_delete_extractions AS
ON DELETE TO extractions_raw DO INSTEAD NOTHING;

CREATE INDEX idx_extractions_case ON extractions_raw(case_id);
CREATE INDEX idx_extractions_module ON extractions_raw(module);
CREATE INDEX idx_extractions_market ON extractions_raw(state, market);
```


facts_labor

Normalized labor rate data — feeds aggregation job

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
extraction_id	UUID	NO	FK → extractions_raw.id
state	TEXT	NO	2-letter state code.
market	TEXT	NO	Metro market.
equipment_type	TEXT	NO	Equipment category.
contract_type	TEXT	YES	Contract scope type.
rate_regular	NUMERIC(10,2)	YES	Regular labor rate per hour (USD).
rate_overtime	NUMERIC(10,2)	YES	Overtime rate per hour (USD).
overtime_multiplier	NUMERIC(5,2)	YES	Overtime as multiple of regular (e.g. 1.5).
travel_minimum_hours	NUMERIC(5,2)	YES	Minimum billable hours per trip.
created_at	TIMESTAMPTZ	NO	Default: NOW()

SQL

```
CREATE TABLE facts_labor (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  extraction_id UUID NOT NULL REFERENCES extractions_raw(id),  
  state TEXT NOT NULL,  
  market TEXT NOT NULL,  
  equipment_type TEXT NOT NULL,  
  contract_type TEXT,  
  rate_regular NUMERIC(10,2),  
  rate_overtime NUMERIC(10,2),  
  overtime_multiplier NUMERIC(5,2),  
  travel_minimum_hours NUMERIC(5,2),  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);  
  
CREATE INDEX idx_labor_market ON facts_labor(state, market, equipment_type);
```

facts_line_items

Invoice line items with taxonomy — Module A

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
extraction_id	UUID	NO	FK → extractions_raw.id

taxonomy_category	TEXT	NO	Canonical component category (e.g. door_operator, controller, hydraulic_unit).
description_normalized	TEXT	YES	Normalized description. Vendor name removed.
amount_billed	NUMERIC(12,2)	YES	Dollar amount on invoice.
labor_hours	NUMERIC(8,2)	YES	Labor hours billed if stated.
scope_flag	TEXT	YES	in_scope out_of_scope ambiguous not_addressed
billing_flag	TEXT	YES	normal anomaly recurring_pattern
created_at	TIMESTAMPZ	NO	Default: NOW()

SQL

```
CREATE TABLE facts_line_items (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  extraction_id UUID NOT NULL REFERENCES extractions_raw(id),  
  taxonomy_category TEXT NOT NULL,  
  description_normalized TEXT,  
  amount_billed NUMERIC(12,2),  
  labor_hours NUMERIC(8,2),  
  scope_flag TEXT,  
  billing_flag TEXT,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);  
  
CREATE INDEX idx_line_items_taxonomy ON facts_line_items(taxonomy_category);
```

facts_parts

Parts and material cost data — Modules A/B

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
extraction_id	UUID	NO	FK → extractions_raw.id
taxonomy_category	TEXT	NO	Canonical component category.
equipment_type	TEXT	NO	Equipment category.
part_cost	NUMERIC(12, 2)	YES	Cost of part(s) as billed.
material_markup_percent	NUMERIC(6, 2)	YES	Markup % applied to materials if stated.
parts_coverage_type	TEXT	YES	included excluded partial not_stated
created_at	TIMESTAMPTZ	NO	Default: NOW()

SQL

```
CREATE TABLE facts_parts (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  extraction_id UUID NOT NULL REFERENCES extractions_raw(id),  
  taxonomy_category TEXT NOT NULL,  
  equipment_type TEXT NOT NULL,  
  part_cost NUMERIC(12,2),  
  material_markup_percent NUMERIC(6,2),  
  parts_coverage_type TEXT,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);  
  
CREATE INDEX idx_parts_taxonomy ON facts_parts(taxonomy_category, equipment_type);
```

facts_contract_terms

Commercial terms — Modules B/C

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
extraction_id	UUID	NO	FK → extractions_raw.id
state	TEXT	NO	State code.
market	TEXT	NO	Metro market.
equipment_type	TEXT	NO	Equipment category.
contract_type	TEXT	YES	Contract scope type.
base_price_per_unit	NUMERIC(12, 2)	YES	Annual base price per elevator unit.

term_years	NUMERIC(4,1)	YES	Contract term in years.
auto_renew_flag	BOOLEAN	YES	TRUE if auto-renew clause present.
escalation_type	TEXT	YES	fixed cpi none not_stated
escalation_percent	NUMERIC(6,3)	YES	Annual escalation % if fixed.
warranty_months	INTEGER	YES	Warranty period in months.
pm_frequency_per_year	INTEGER	YES	Preventive maintenance visits per year.
after_hours_policy	TEXT	YES	included billed_separately not_addressed
created_at	TIMESTAMPTZ	NO	Default: NOW()

SQL

```
CREATE TABLE facts_contract_terms (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  extraction_id UUID NOT NULL REFERENCES extractions_raw(id),
  state TEXT NOT NULL,
  market TEXT NOT NULL,
  equipment_type TEXT NOT NULL,
  contract_type TEXT,
  base_price_per_unit NUMERIC(12,2),
  term_years NUMERIC(4,1),
  auto_renew_flag BOOLEAN,
  escalation_type TEXT,
  escalation_percent NUMERIC(6,3),
  warranty_months INTEGER,
  pm_frequency_per_year INTEGER,
  after_hours_policy TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_contract_market ON facts_contract_terms(state, market, equipment_type);
```

benchmarks

Versioned aggregated intelligence — the flywheel output

This table is written by the nightly aggregation job — never manually. published = FALSE when sample_size < 25. Only published rows are injected into AI prompts.

Column	Type	Null?	Notes
id	UUID	NO	Primary key.
benchmark_version	TEXT	NO	Semantic version (e.g. '1.2'). Increments each aggregation run.
dimension	TEXT	NO	What is being benchmarked: labor_regular overtime_multiplier base_price_per_unit escalation_percent material_markup_percent warranty_months
state	TEXT	NO	State code.
market	TEXT	NO	Metro market.
equipment_type	TEXT	NO	Equipment category.
contract_type	TEXT	YES	Optional filter dimension.
sample_size	INTEGER	NO	Number of data points used. Must be >= 25 to publish.
median_value	NUMERIC(12,4)	YES	Median value across sample.
p25_value	NUMERIC(12,4)	YES	25th percentile.
p75_value	NUMERIC(12,4)	YES	75th percentile.
mean_value	NUMERIC(12,4)	YES	Mean value.
stddev_value	NUMERIC(12,4)	YES	Standard deviation.
published	BOOLEAN	NO	Default: FALSE. Set TRUE by aggregation job when sample_size >= 25.
aggregated_at	TIMESTAMPTZ	NO	When this benchmark row was calculated.

SQL

```
CREATE TABLE benchmarks (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  benchmark_version TEXT NOT NULL,  
  dimension TEXT NOT NULL,  
  state TEXT NOT NULL,  
  market TEXT NOT NULL,  
  equipment_type TEXT NOT NULL,  
  contract_type TEXT,  
  sample_size INTEGER NOT NULL,  
  median_value NUMERIC(12,4),  
  p25_value NUMERIC(12,4),
```

```
p75_value NUMERIC(12,4),
mean_value NUMERIC(12,4),
stddev_value NUMERIC(12,4),
published BOOLEAN NOT NULL DEFAULT FALSE,
aggregated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_benchmarks_lookup
ON benchmarks(state, market, equipment_type, dimension, published);
CREATE INDEX idx_benchmarks_version ON benchmarks(benchmark_version);
```

Benchmark Thresholds

Condition	Action
sample_size >= 25	published = TRUE — included in retrieval injection
sample_size < 25	published = FALSE — excluded from AI prompts
sample_size < 10	Do not aggregate — wait for more data
value > 3x IQR	Exclude from aggregation — log to outlier_log

Anonymization Rules — Required Before Any DB Insert

These rules are mandatory. Violation exposes customer data and corrupts benchmark integrity.